

1. (Fall 2008, #3) You want to approximate the integral $\int_1^2 e^{x^2} dx$ by using the Trapezoidal Rule with n intervals. How large should you take n to guarantee that the error is at most 10^{-6} ? (Do not compute the approximation.)

3 ERROR BOUNDS Suppose $|f''(x)| \leq K$ for $a \leq x \leq b$. If E_T and E_M are the errors in the Trapezoidal and Midpoint Rules, then

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$\begin{aligned} f(x) &= e^{x^2} & f'(x) &= e^{x^2} \cdot 2x & f''(x) &= e^{x^2} \cdot 2x \cdot 2x \\ & & & & & + e^{x^2} \cdot 2 \\ & & & & & = e^{x^2} (4x^2 + 2) \end{aligned}$$

$$\text{for } x \in [1, 2]$$

$$\begin{aligned} |f''(x)| &= |e^{x^2} (4x^2 + 2)| \leq e^{2^2} \cdot (4 \cdot 2^2 + 2) = e^4 \cdot 18 \\ \Rightarrow K &= 18e^4. \end{aligned}$$

$$|E_T| \leq \frac{18e^4 \cdot (2-1)^3}{12n^2} \leq 10^{-6}$$

$$\Rightarrow n^2 \geq \frac{18e^4}{12} \cdot 10^6,$$

$$\Rightarrow n \geq \underbrace{\sqrt{\frac{3e^4}{2}}}_{\approx 9.0497} \cdot 10^3.$$

$$n \geq 9049.7 \quad \text{choose } n = 9050.$$

2. (Fall 2008, #3) We want to approximate the value of the integral $\int_1^3 \frac{1}{x^2} dx$ by using the Midpoint Rule and subdividing the $[1, 3]$ into n intervals of equal length. How large should we take n if we want the error to be less than 0.0002?

$$f(x) = \frac{1}{x^2} = x^{-2} \quad f'(x) = -2x^{-3} \quad f''(x) = (-2) \cdot (-3)x^{-4} = 6x^{-4}.$$

$$|f''(x)| = |6x^{-4}| \leq 6 \quad \text{for } x \in [1, 3],$$

Choose $K = 6$.

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$|E_M| \leq \frac{6(3-1)^3}{24n^2} \leq 0.0002.$$

$$\Rightarrow n^2 \geq \frac{6 \times 8}{24 \times 2 \times 10^{-4}} = 10^4;$$

$$\Rightarrow n \geq 100.$$

3. (Stewart, §6.5, 31)

- (a) Use the Midpoint Rule and the given data to estimate the value of the integral $\int_1^5 f(x)dx$.

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
$f(x)$	2.4	2.9	3.3	3.6	3.8	4.0	4.1	3.9	3.5

MIDPOINT RULE

$$\int_a^b f(x) dx \approx M_n = \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)]$$

where $\Delta x = \frac{b-a}{n}$

and $\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i]$

$$x_0 = 1 \quad x_1 = 2 \quad x_2 = 3 \quad x_3 = 4 \quad x_4 = 5$$

$$\bar{x}_1 = 1.5 \quad \bar{x}_2 = 2.5 \quad \bar{x}_3 = 3.5 \quad \bar{x}_4 = 4.5$$

$$n = 4. \quad \Delta x = \frac{5-1}{4} = 1.$$

$$M_4 = 1 \cdot [f(1.5) + f(2.5) + f(3.5) + f(4.5)] \\ = 2.9 + 3.6 + 4.0 + 3.9 = 14.4$$

- (b) If it is known that $-2 \leq f''(x) \leq 3$ for all x , estimate the error involved in the approximation in part (a).

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$K = 3. \quad |E_M| \leq \frac{3 \cdot (5-1)^3}{24 \cdot 4^2} = 0.5.$$

SIMPSON'S RULE

$$\int_a^b f(x) dx \approx S_n = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

where n is even and $\Delta x = (b - a)/n$.

4. (Stewart, §6.5, 30)

(a) A table of values of a function g is given. Use Simpson's Rule to estimate $\int_0^{1.6} g(x) dx$.

x	x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
$g(x)$	12.1	11.6	11.3	11.1	11.7	12.2	12.6	13.0	13.2

$$n=8 \quad \Delta x = \frac{1.6}{8} = 0.2$$

$$S_8 = \frac{0.2}{3} [12.1 + 4 \times 11.6 + 2 \times 11.3 + 4 \times 11.1 + 2 \times 11.7 + 4 \times 12.2 + 2 \times 12.6 + 4 \times 13.0 + 13.2]$$

(b) If $-5 \leq g^{(4)}(x) \leq 2$ for $0 \leq x \leq 1.6$, estimate the error involved in the approximation in part (a).

4 ERROR BOUND FOR SIMPSON'S RULE Suppose that $|f^{(4)}(x)| \leq K$ for $a \leq x \leq b$. If E_S is the error involved in using Simpson's Rule, then

$$|E_S| \leq \frac{K(b-a)^5}{180n^4}$$

$$K=5 \quad |E_S| \leq \frac{5 \times (1.6-0)^5}{180 \times 8^4}$$

Note: For Simpson's rule, n must be even.

5. (Stewart, §6.5, 17)

(a) Find the approximations T_8 and M_8 for the integral $\int_0^1 \cos(x^2) dx$.

TRAPEZOIDAL RULE

$$\int_a^b f(x) dx \approx T_n = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)]$$

where $\Delta x = (b - a)/n$ and $x_i = a + i \Delta x$.

$$\Delta x = \frac{1-0}{8} = \frac{1}{8}$$

$$x_0 = 0 \quad x_1 = \frac{1}{8} \quad x_2 = \frac{2}{8} \quad \cdots \quad x_8 = 1.$$

$$T_8 = \frac{1}{2} \times \frac{1}{8} \times [\cos(0) + 2\cos((\frac{1}{8})^2) + 2\cos((\frac{2}{8})^2) + \cdots + 2\cos((\frac{7}{8})^2) + \cos(1)].$$

MIDPOINT RULE

$$\int_a^b f(x) dx \approx M_n = \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)]$$

where $\Delta x = \frac{b-a}{n}$

and $\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i]$

$$\Delta x = \frac{1}{8} \quad \bar{x}_1 = \frac{1}{16} \quad \bar{x}_2 = \frac{3}{16}, \bar{x}_3 = \frac{5}{16}, \cdots \quad \bar{x}_8 = \frac{15}{16}$$

$$M_8 = \frac{1}{8} [\cos((\frac{1}{16})^2) + \cos((\frac{3}{16})^2) + \cdots + \cos((\frac{15}{16})^2)].$$

(b) Estimate the errors in the approximations of part (a).

3 ERROR BOUNDS Suppose $|f''(x)| \leq K$ for $a \leq x \leq b$. If E_T and E_M are the errors in the Trapezoidal and Midpoint Rules, then

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$f(x) = \cos(x^2) \quad f'(x) = -\sin(x^2) \cdot 2x$$

$$f''(x) = -\cos(x^2) \cdot 2x \cdot 2x - \sin(x^2) \cdot 2$$

$$= -\cos(x^2) \cdot 4x^2 - \sin(x^2) \cdot 2 \quad x \in [0, 1]$$

$$\Rightarrow |f''(x)| \leq 4 + 2 = 6 =: K.$$

(c) How large do we have to choose n so that the approximations T_n and M_n to the integral in part (a) are accurate to within 0.0001?

$$\Rightarrow |E_T| \leq \frac{6 \cdot (1-0)^3}{12 \times 8^2}$$

$$|E_M| \leq \frac{6 \times (1-0)^3}{24 \times 8^2}$$

$$|E_T| \leq \frac{6 \cdot 1}{12n^2} \leq 10^{-4}$$

$$\Rightarrow n^2 \geq \frac{1}{2} \times 10^4 = 5000$$

$$\Rightarrow n \geq \sqrt{5000} = 70.71 \quad \text{choose } n = 71.$$

$$|E_M| \leq \frac{6 \times 1}{24 \times n^2} \leq 10^{-4}$$

$$\Rightarrow n^2 \geq \frac{1}{4} \times 10^4$$

$$\Rightarrow n \geq 50.$$